

Indian Statistical Institute

End Semester Examination (Part II)

Optimization and Numerical Methods
20 May 2026, 4:00 pm – 5:30 pm

Bachelor of Statistical Data Science
Maximum marks in Part II: 27

INSTRUCTIONS

- The exam is divided into two parts: Part I and Part II. Each part carries a maximum of 27 marks. However, note that the total marks that can be scored for this exam, combining both the parts, are capped at 50.
- Part II contains three questions. Answer ALL the questions. The marks assigned to each question are indicated alongside the question.
- Use of scientific calculator is permitted. No other electronic devices are allowed in the examination hall.

QUESTIONS

- (a) Let $f : [0, 1] \rightarrow \mathbb{R}$ be defined as $f(x) = x^2 + 1$. Let I_n denote the approximation of $\int_0^1 f(x)dx$ using the trapezoidal rule with n equal sub-intervals, i.e., with nodes $0 = x_0 < x_1 < \dots < x_n = 1$, where $x_k = \frac{k}{n}$ for all $0 \leq k \leq n$.
 - (i) Show that $I_n = 1 + \frac{1}{2n} + \frac{(n-1)(2n-1)}{6n^2}$, for every $n \geq 1$. [4]
 - (ii) Compute the actual value $I = \int_0^1 f(x)dx$ analytically, and show that $I_n \rightarrow I$ as $n \rightarrow \infty$. [3]
- (b) Let X be a random variable with probability density function $f(x) = \frac{2x}{3}$, for $1 \leq x \leq 2$. Compute the expected value $\mathbb{E}[e^X]$ using Simpson's 1/3 rule, with $n = 4$ sub-intervals. Carry at least four decimal places throughout the calculations. [2]
- (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function. Consider the tangent line to the curve $y = f(x)$ at the point $(x_k, f(x_k))$. By finding the x -intercept of the line, derive the Newton-Raphson iteration formula: [2]

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

- (b) In this problem, we will write an algorithm to compute the reciprocal ($1/A$) of a given positive real number A using the Newton-Raphson method.
 - (i) Consider the function $f(x) = \frac{1}{x} - A$. Show that the Newton-Raphson iteration becomes $x_{k+1} = x_k(2 - Ax_k)$, $k \geq 0$. [2]
 - (ii) Using this iteration, write an algorithm that: [3]

- Takes as inputs A (the number), ε (error tolerance), and an initial guess x_0 .
 - Computes the current approximation x_{k+1} and the error $|x_{k+1} - x_k|$, in every iteration.
 - Stops when the error is less than ε and returns the final value.
- (c) Using the algorithm, compute the reciprocal of $A = 1.75$, with inputs $\varepsilon = 0.01$ and $x_0 = 0.5$. Carry at least four decimal places throughout the calculations. [2]
3. (a) Prove that any locally optimal point of a convex optimization problem is also globally optimal. [4]
- (b) Consider a binary random variable y_i which only takes values 0 and 1, and predictor variable $x_i, i = 1, \dots, n$. Also assume the binary logistic regression model as follows

$$p_i = P(y_i = 1 \mid x_i) = \frac{e^{x_i \beta}}{1 + e^{x_i \beta}},$$

where β is unknown regression coefficient. Suppose the task is to compute the maximum likelihood estimate (MLE) of β based on a sample of size n . Derive the likelihood function of β . And write the step-by-step algorithm to compute the MLE using gradient descent method. [2+3]